



EPID Image Prediction for Pretreatment QA Using Generative Deep Learning

: Impact of Label Data Variations and Daily Machine Variability

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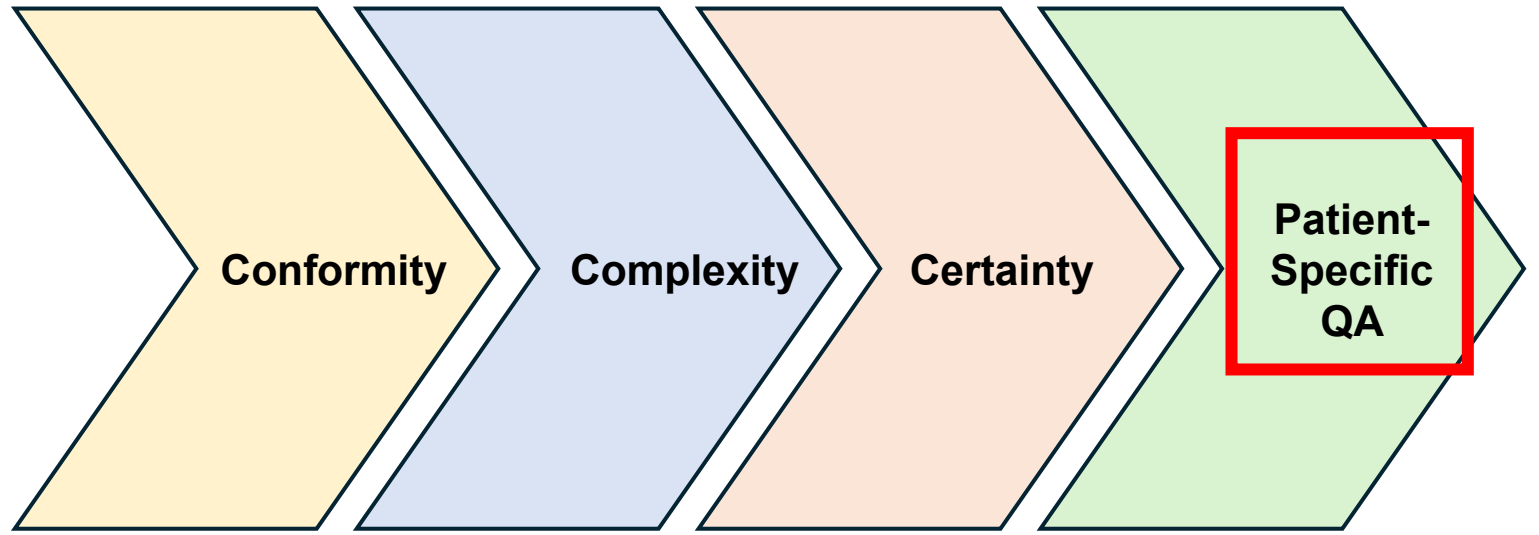
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Introduction

Introduction

Patient-specific Pretreatment QA (PSQA)



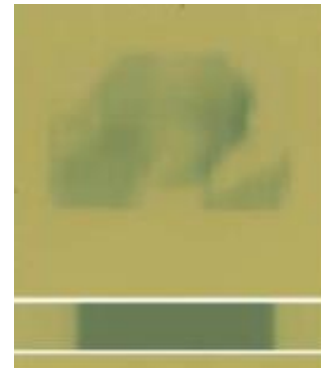
**PSQA
methodology**



Electric Portal Imaging
Device (EPID)



PSQA Phantom



Radiochromic film



Phantom & Ion chamber

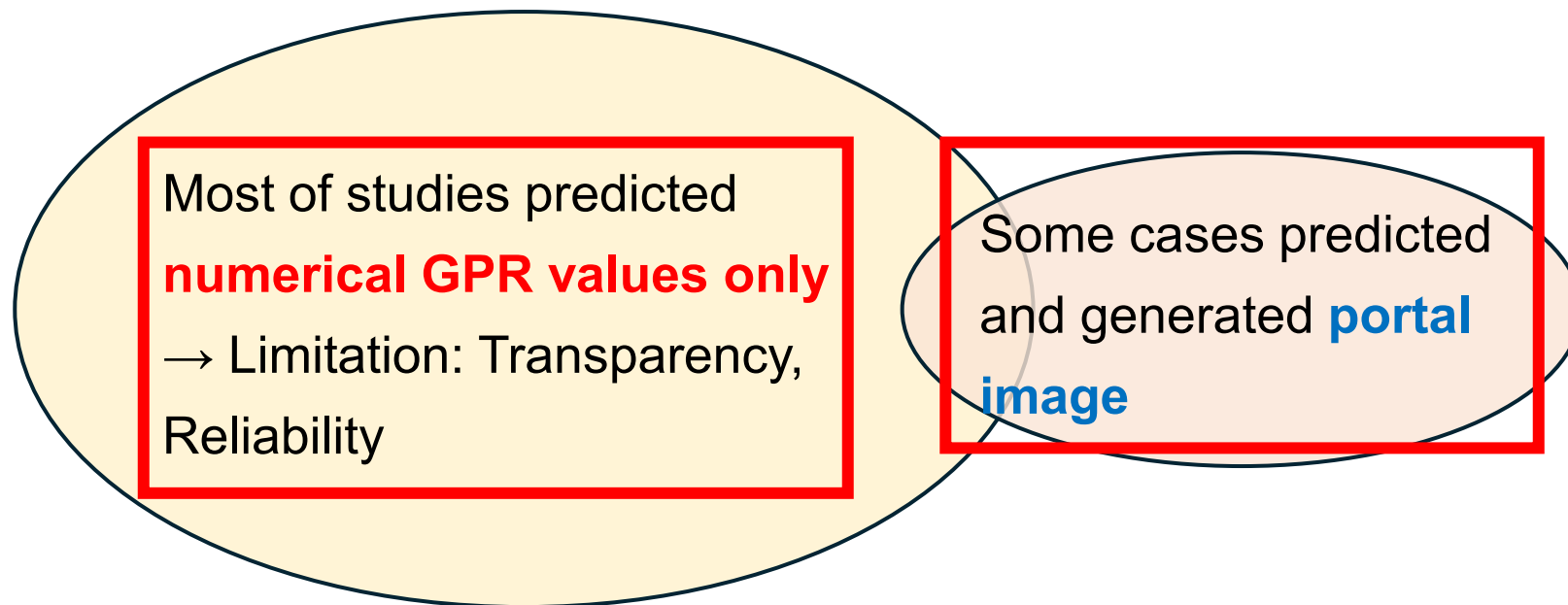
- Patient-Specific pre-treatment QA is essential to IMRT or VMAT plans (ICRU 83, AAPM TG-218)



Introduction

Approach to apply of ML/DL to PSQA

- Approaches to **applying Machine Learning/Deep Learning to PSQA** have spread.
→ Some studies refer to this approach as **'Virtual PSQA'**.
- **Predict the Gamma Passing Rate (GPR)** based on plan parameters, MLC sequence maps, EPID images, etc.





Introduction

EPID Image Prediction for PSQA using Deep learning algorithm

Deep Learning
algorithm to
PSQA

Transparency,
Reliability

Direct application to the
current analysis method

Portal Image
Prediction

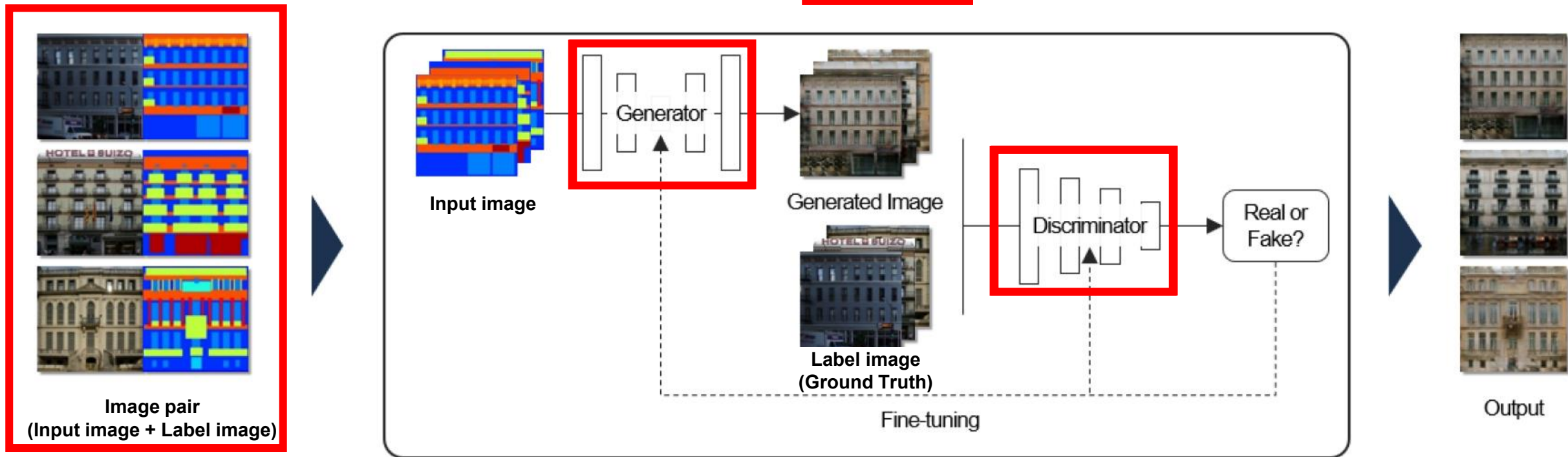
Generative
Adversarial
Network (GAN)
model

- 1) **Predicted image generation** → Comparable with **current Gamma evaluation** method.
- 2) The impact of **label data shape variations** was evaluated.
- 3) Machine Performance Check (MPC) data was included to consider **daily machine status**.

Materials and Methods

Material and Methods

Generative Adversarial Network – Pix2Pix

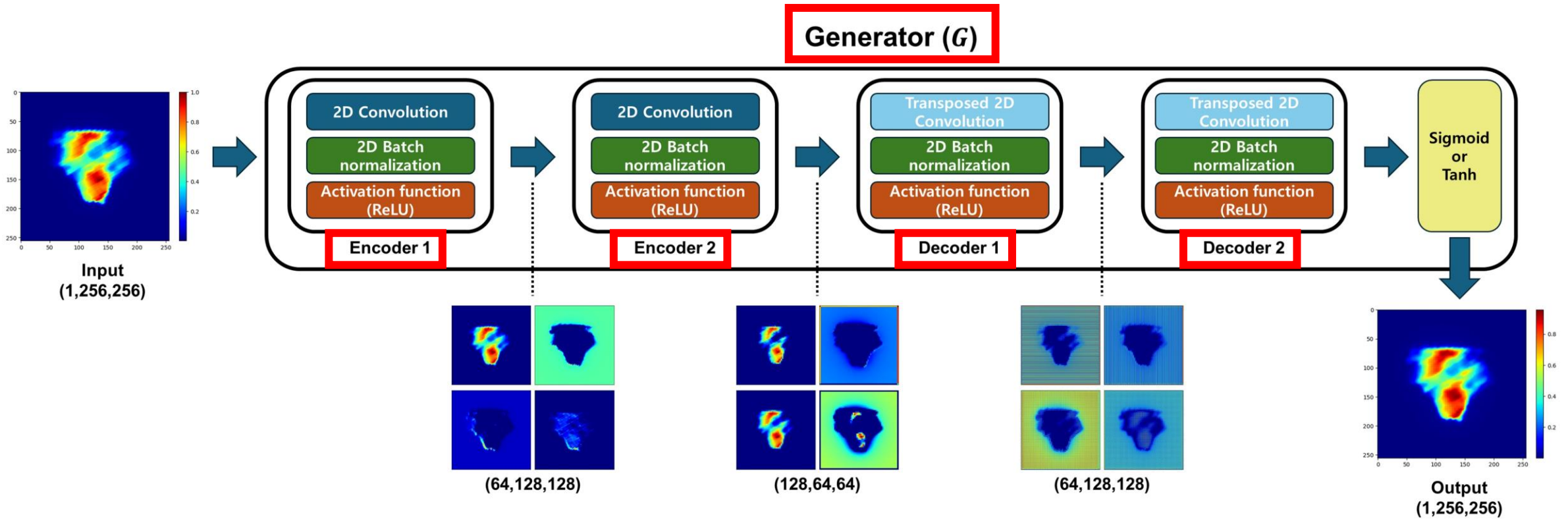


- Specialty in **Image-to-Image translation**
- The **generator** and **discriminator** learn in a complementary (adversarial) manner.
→ improves the performance of the generator.
- The discriminator outputs **real [1]** or **fake [0]** (average output 0.5 → an ideal balance).

Material and Methods

Generative Adversarial Network – Pix2Pix

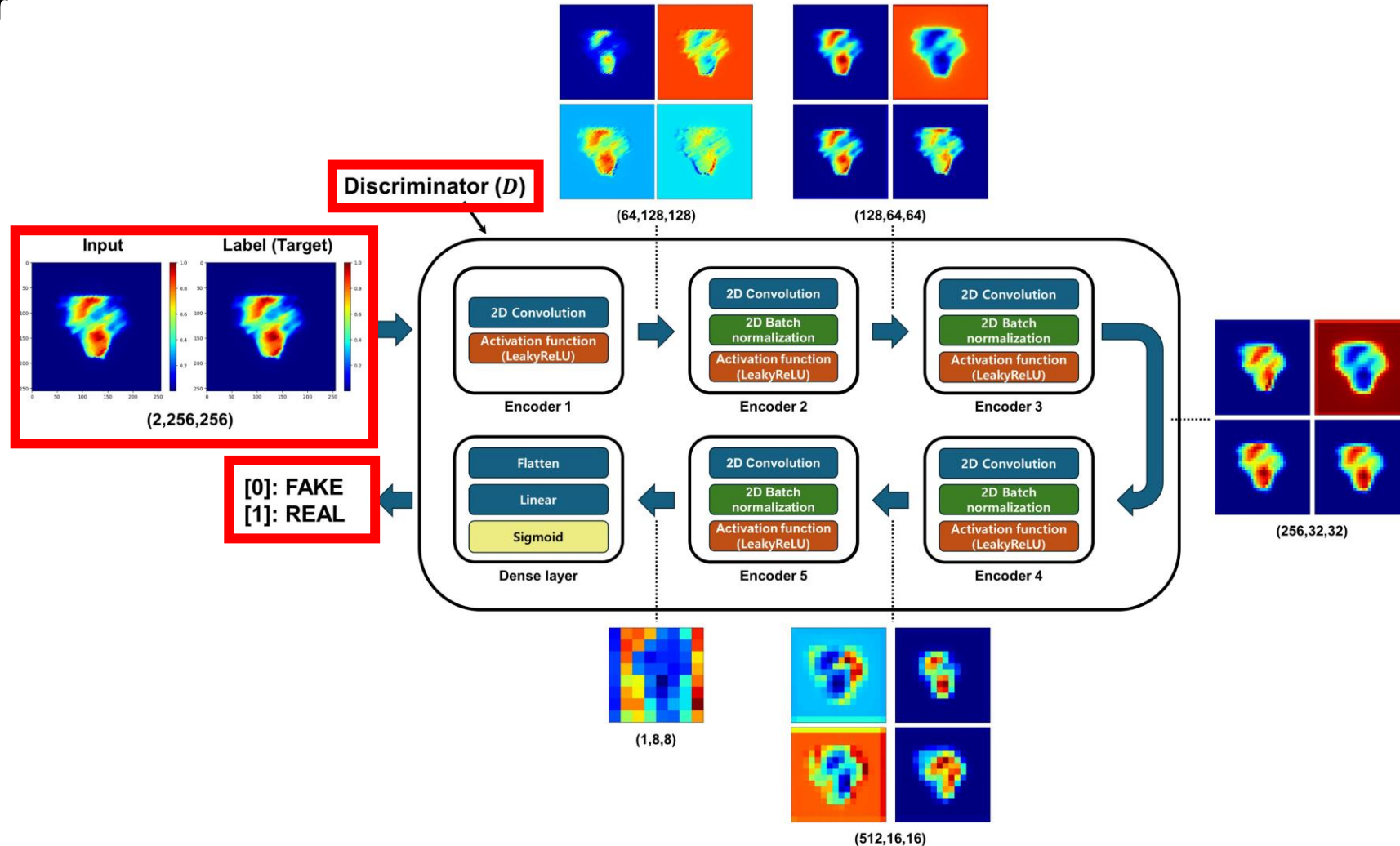
Generator



Material and Methods

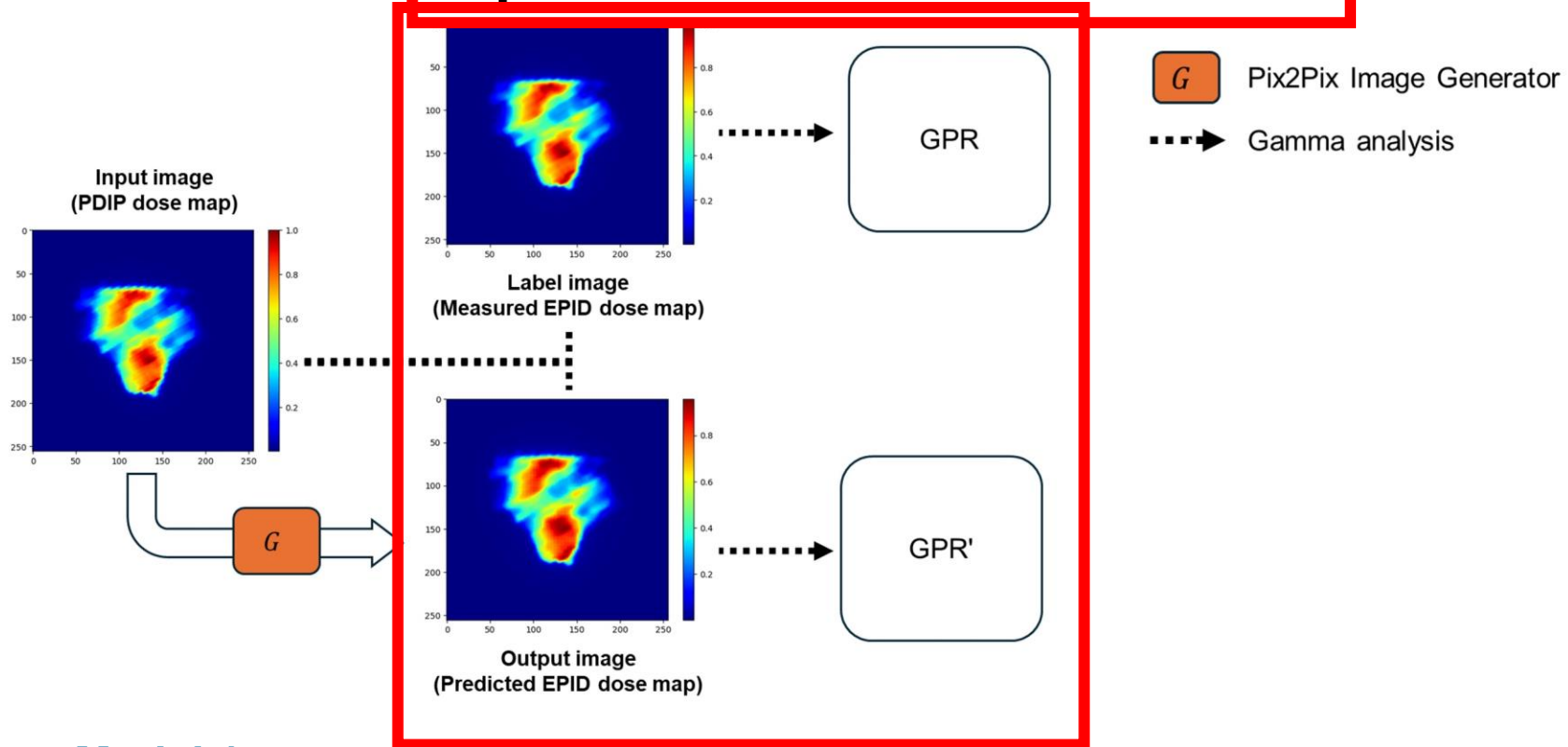
Generative Adversarial Network – Pix2Pix

Discriminator



Material and Methods

Model 1 vs Model 2 - Impact of Label Data Variations

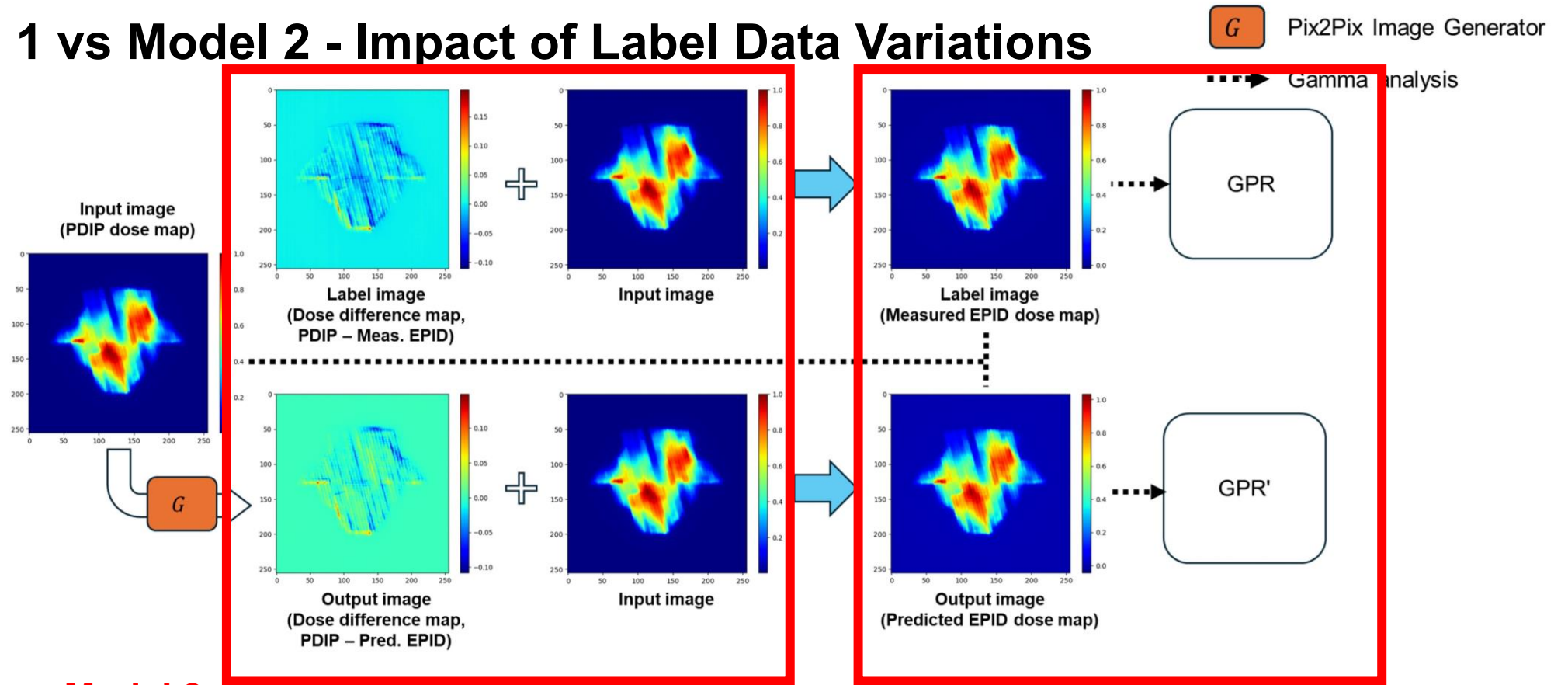


- Model 1

- Input: The fluence map calculated in TPS using AAA algorithm.
- Label: The fluence map measured by EPID
- Output: The EPID fluence map predicted by Pix2Pix model

Material and Methods

Model 1 vs Model 2 - Impact of Label Data Variations



- Model 2

- Input: The fluence map calculated in TPS using AAA algorithm.
- Label: The dose difference map (calculated map – measured map)
- Output: The dose difference map predicted by Pix2Pix model



Material and Methods

※ The use of portal image data has been approved by the Institutional Review Board of Asan Medical Center (IRB No. 2025-0870)

Development phase 1 – Basic architecture verification

Training Prostate 61 fields / **Test** Prostate 13 fields (Varian Halcyon)

Development phase 2 – Generalization through various treatment sites

Training H&N 16, Thorax 12, Pelvis 73, Others 6 fields / **Test** H&N 5, Thorax 2, Pelvis 22 fields

Development phase 3 – Including daily Machine status

Training H&N 16, Thorax 12, Pelvis 73, Others 6 fields / **Test** H&N 5, Thorax 2, Pelvis 22 fields

MPC data at measurement date included to the input tensor

→ MLC leaf positioning errors (231 features)

- Performance evaluation: Gamma index analysis (3%/3 mm – 2%/2 mm)
→ Mean Absolute Error (MAE), Pearson's correlation coefficient (r)



Material and Methods

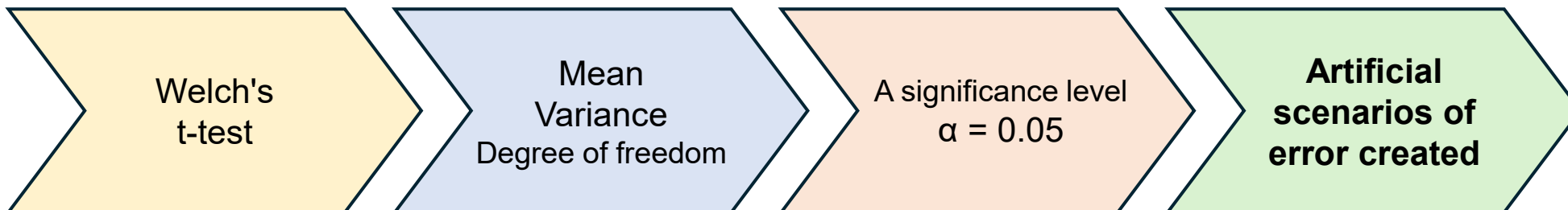
Statistical verification (*p-value*)

- To assess the practicality of incorporating daily machine status into performance improvement
- Null hypothesis

No significant difference in prediction performance between the model utilizing MPC data and the model without it

- Alternative hypothesis

A model utilizing MPC data would demonstrate superior predictive performance

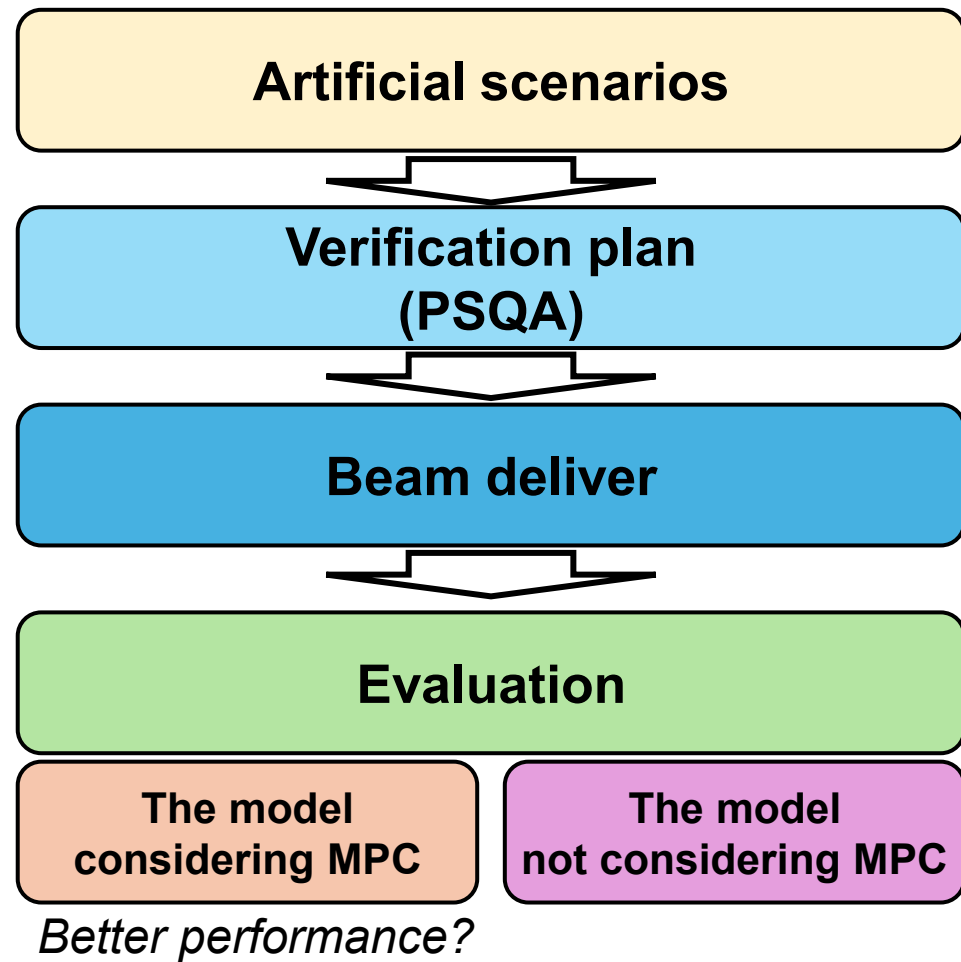


Material and Methods

Statistical verification (*p-value*)

- Artificial scenarios simulating MLC positioning errors were created.

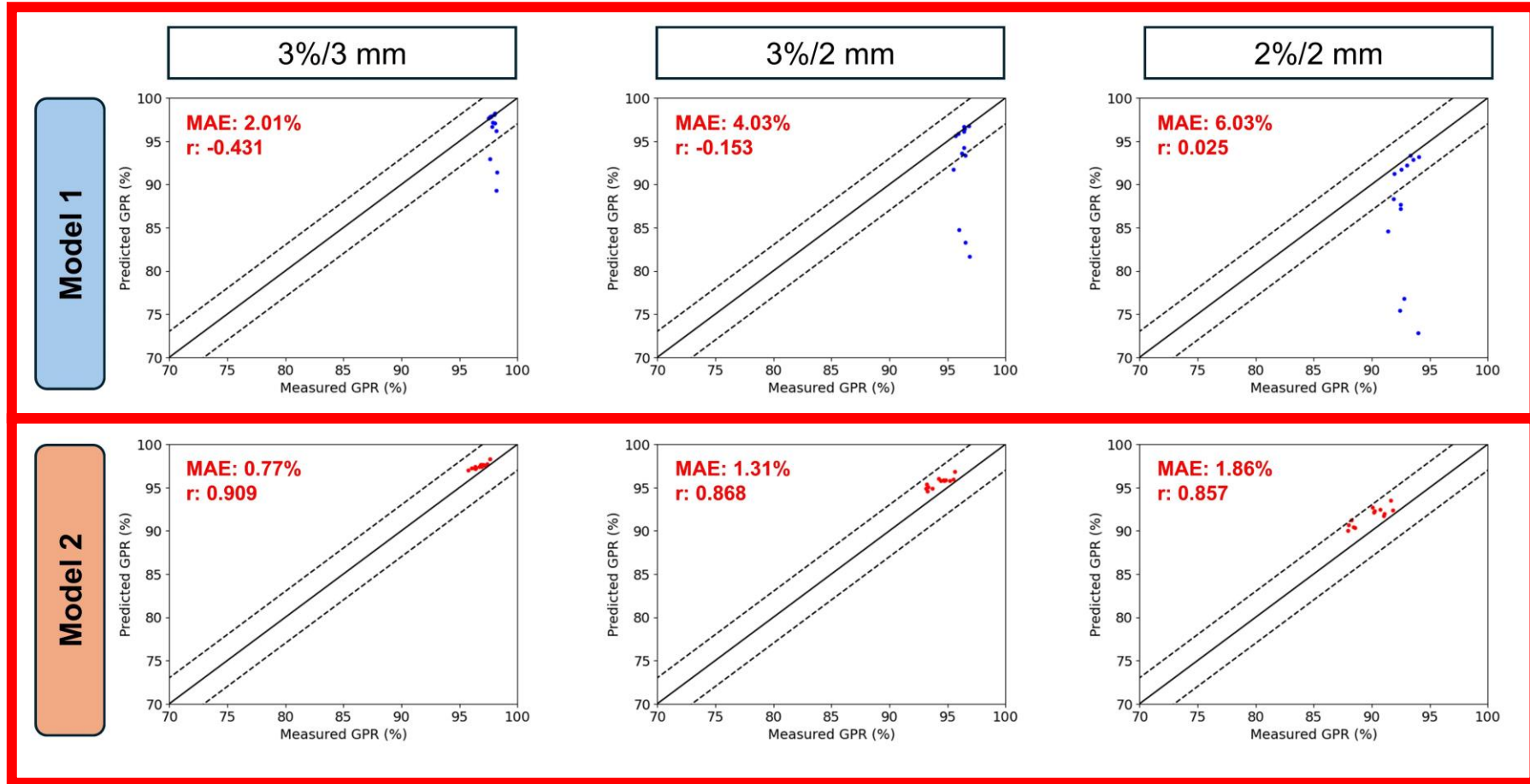
MLC adjustment pattern No.	Bank A		Bank B	
	Adjusted leaves	Offset (mm)	Adjusted leaves	Offset (mm)
1	All	0	All	-1
2	All	+1	All	0
3	All	+1	All	+1
4	All	-1	All	+1
5	12, 13, 17, 40, 41, 45	-1	12, 13, 17, 40, 41, 45	+1
6	12, 13, 17, 40, 41, 45	+1	12, 13, 17, 40, 41, 45	-1



Results

Result

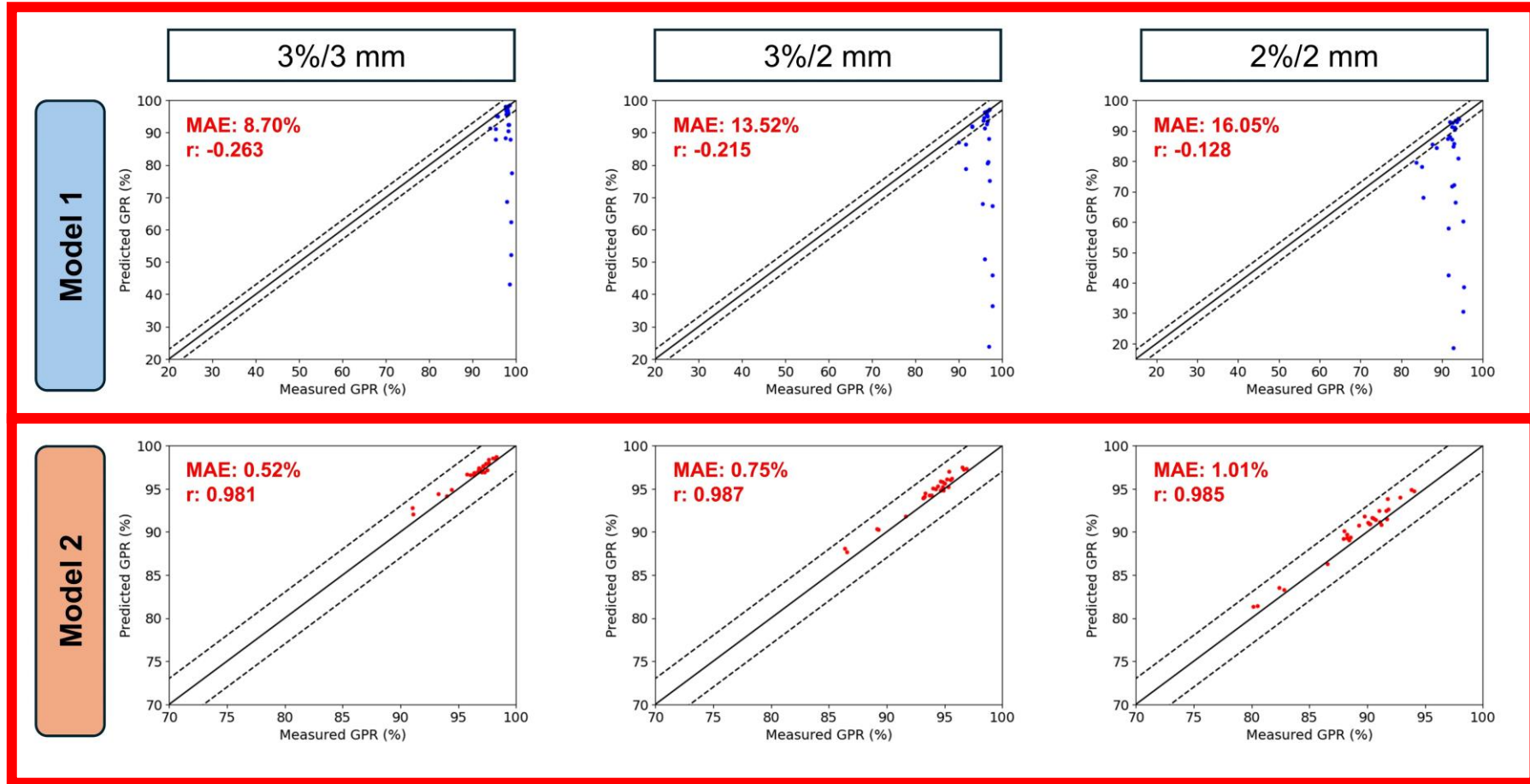
Development phase 1 - Base architecture verification



- Treatment site: Limited to prostate

Result

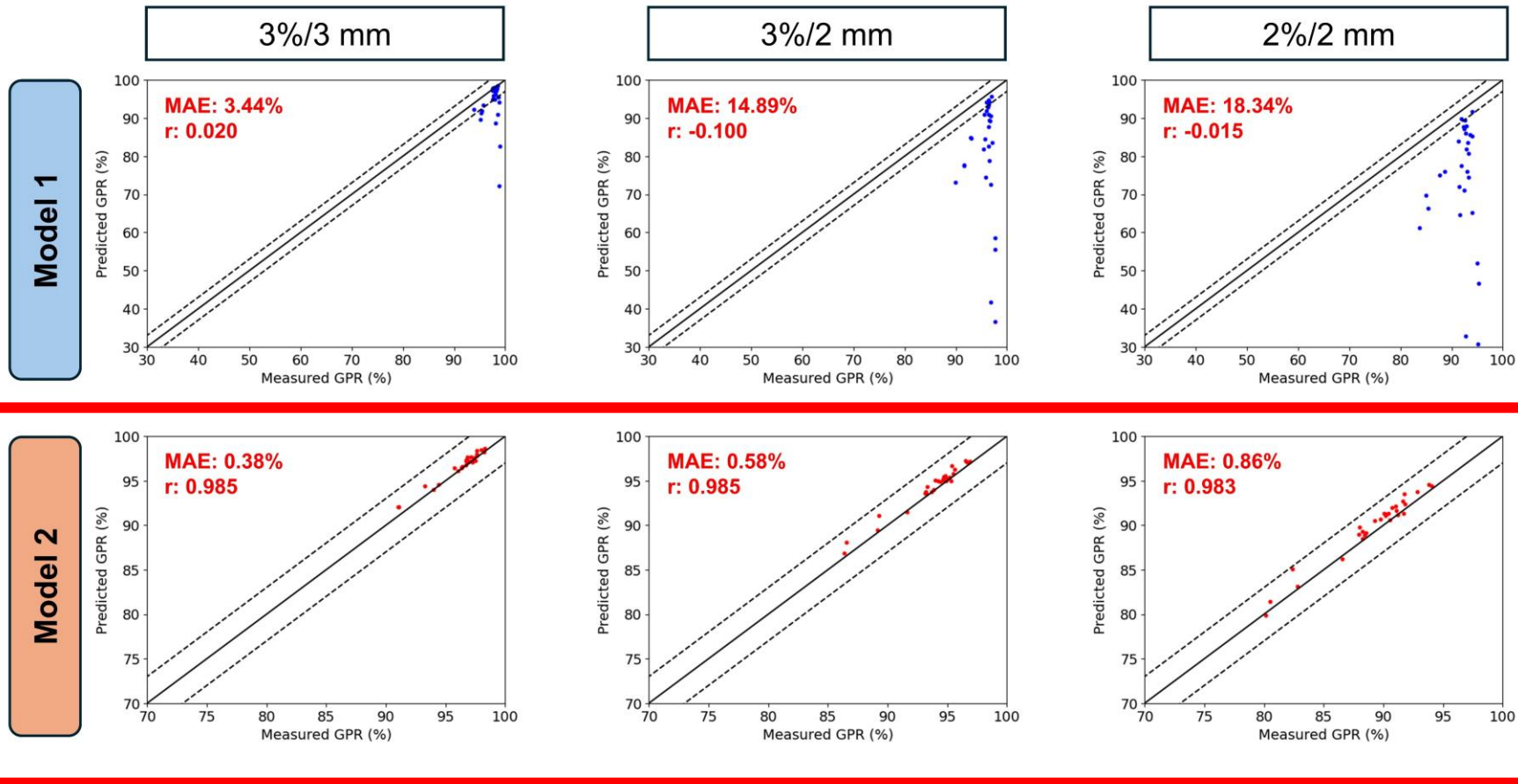
Development phase 2 – Generalization over treatment sites



- Treatment sites: HN, Thorax, Abdomen, Pelvis

Result

Development phase 3 - Including daily Machine status



- Treatment sites: HN, Thorax, Abdomen, Pelvis
- MPC data included in the input data



Result

Summary

Model 1		3%/3mm	3%/2mm	2%/2mm
MAE (Pearson's r)	Phase1	2.01% (-0.431)	4.03% (-0.153)	6.03% (0.025)
	Phase2	8.7% (-0.263)	13.52% (-0.215)	16.05% (-0.128)
	Phase3	3.44% (0.020)	14.89% (-0.100)	18.34% (-0.015)

Model 2		3%/3mm	3%/2mm	2%/2mm
MAE (Pearson's r)	Phase1	0.77% (0.909)	1.31% (0.868)	1.86% (0.857)
	Phase2	0.52% (0.981)	0.75% (0.987)	1.01% (0.985)
	Phase3	0.38% (0.985)	0.58% (0.985)	0.86% (0.983)

Result

Statistical verification (p-value)

Plan No.		1	2	3	4	5	6	7	8	9	10
Ground truth	$X_{i,GT}$	98.23	98.12	98.25	98.07	93.58	92.95	93.29	93	92.88	91.95
	$X_{i,MPC}$	98.42	98.35	98.45	98.29	93.86	93.53	93.55	93.51	91.79	91.86
The model considering MPC	$ X_{i,MPC} - X_{i,GT} $	0.19	0.23	0.2	0.22	0.28	0.58	0.26	0.51	1.09	0.09
	MAE_{MPC}	0.365									
	S_{MPC}^2	0.087									
The model not considering MPC	$X_{i,Gen.}$	98.45	98.58	98.45	98.58	94.17	94.37	94.17	94.37	92.03	92.8
	$ X_{i,Gen.} - X_{i,GT} $	0.22	0.46	0.2	0.51	0.59	1.42	0.88	1.37	0.85	0.85
	$MAE_{Gen.}$	0.735									
	$S_{Gen.}^2$	0.180									
	T-statistic	-2.265									
	Degree of freedom (ν)	16.1									

- p -value (Two-sided): **0.038** ($< \alpha$ of 0.05)

Conclusion and Discussion



Conclusion and Discussion

Conclusion and Discussion

- The proposed **GAN-based model** enables accurate **prediction of EPID fluence maps**.
- The prediction performance was compared based on **the label data shape**.
 - Fluence map vs Dose difference map
- **The integration of real-time machine performance data**, enhancing robustness and clinical relevance.
 - statistically significant verification completed (p-value).
- This framework provides a foundation for **hybrid PSQA strategies**.
 - reduce measurement workload and improve efficiency in routine radiotherapy QA workflows.



Conclusion and Discussion

Future Works

- Validation and implementation of clinical applicability after peer review.
- Expansion of training data and refinement of the model.

Generative Adversarial Network-Based Prediction of EPID Dose Maps for Patient-Specific Quality Assurance: Incorporating Label Variation and Daily Machine Variability

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Abstract Patient-specific quality assurance is essential in intensity-modulated and volumetric modulated arc radiotherapy, but conventional methods using electronic portal imaging device (EPID) measurements are resource-intensive and accelerate aging of the imaging detector. We present a deep learning framework based on a conditional Generative Adversarial Network (GAN) for predicting EPID dose maps, aiming to reduce reliance on physical measurements while ensuring clinical accuracy. Built on the Pix2Pix architecture, the model was trained on 136 VMAT treatment fields with two label data types: direct EPID dose maps and dose difference maps. To incorporate daily machine variability, 231 machine performance check (MPC) parameters were included in the input. Evaluation using gamma passing rates, mean absolute error (MAE), and Pearson’s correlation showed that the dose difference model outperformed direct regression, achieving MAE of $0.58\% \pm 0.45$ and correlation of 0.985 under a 3%/2 mm gamma criterion. The model generalized well across various treatment sites and

**Undergoing peer review
(Scientific Reports, IF 3.9)**

발명신고서

1. 발명의 명칭		
방사선기기의 실시간 상태를 반영한 생성형 인공지능 기반 EPID 이미지 예측을 이용한 세기변조방사선치료 환자별 정도관리 방법		
2. 발명 정보 (해당되는 항목에 v 표시)		
발명의 속성	a. 새로운 개념을 제시한 기본발명이다. () b. 기존기술을 대체하는 대안을 제시한 발명이다. (v) c. 기존기술의 개량한 발명이다. ()	
	근거	방사선치료 환자별 정도관리 방법 중 EPID(Electronic Portal Imaging Device) 측정 방식의 인적/시간적 소모를 보완하기 위하여, 방사선기기의 실시간 상태를 반영하는 적대적 생성형 네트워크 등 인공지능모델을 활용한 EPID 이미지 예측을 통한 방사선치료의 환자별 정도관리 방법을 제시함.
		a. 시제품 상용화 단계 () b. 시제품 제작 단계 ()

**Patent application in progress
(AMC R&D team)**

Journal Review

PHYSICS CONTRIBUTION

Prospective Clinical Validation of Virtual Patient-Specific Quality Assurance of Volumetric Modulated Arc Therapy Radiation Therapy Plans



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Received Jul 20, 2021; Accepted for publication Apr 27, 2022

Purpose: Performing measurement-based patient-specific quality assurance (PSQA) is recognized as a resource-intensive and time inefficient task in the radiation therapy treatment workflow. Paired with technological refinements in modern radiation therapy, research toward measurement-free PSQA has seen increased interest during the past 5 years. However, these efforts have not been clinically implemented or prospectively validated in the United States. We propose a virtual QA (VQA) system and workflow to assess the safety and workload reduction of measurement-free PSQA.

- Prospective clinical validation of virtual PSQA
- Machine learning model (XGBoost) to predict **the difference** between ion chamber measurement and TPS dose. (numerical value)



Purpose

- Measurement-based Patient-Specific QA of VMAT plans is a **time- and resource-intensive task**.
- Research toward **measurement-free PSQA** has seen increased interest **since 2017**.
→ Boomed with ML/DL development
- The first study to demonstrate **clinical implementaion** on large **prospective** validation data.
- **To assess the safety and workload reduction** of measurement-free PSQA.

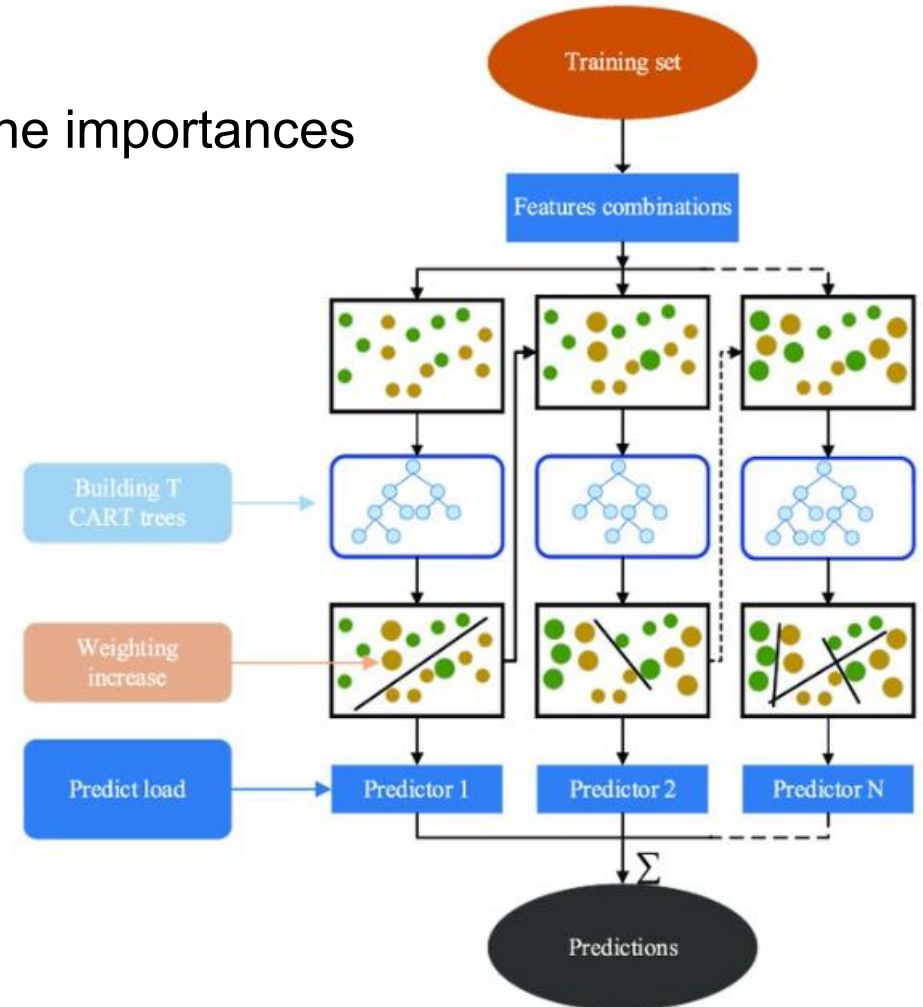
Methods and Materials

- Prediction model: **XGBoost** (ML, Decision Tree)
 - Scoring importance of each feature and predict based the importances
- Complexity features of plans (parameters for prediction)

Feature
FFF beam model
Edge metric
Modulation complexity score
MU factor
Dose rate
Collimator angle
Jaw plan irregularity

...

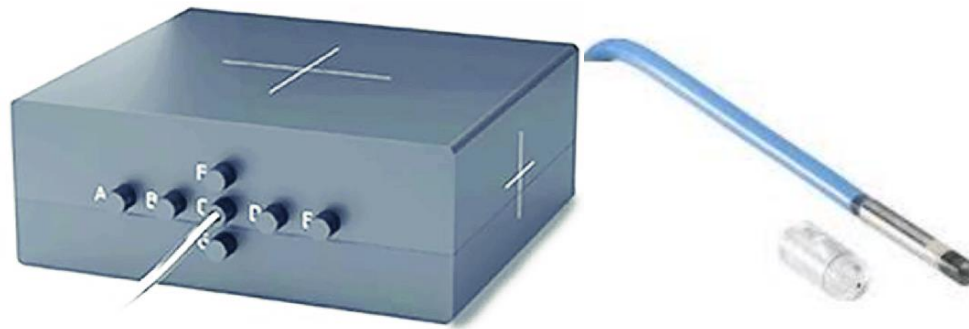
→ Totally 293 raw features



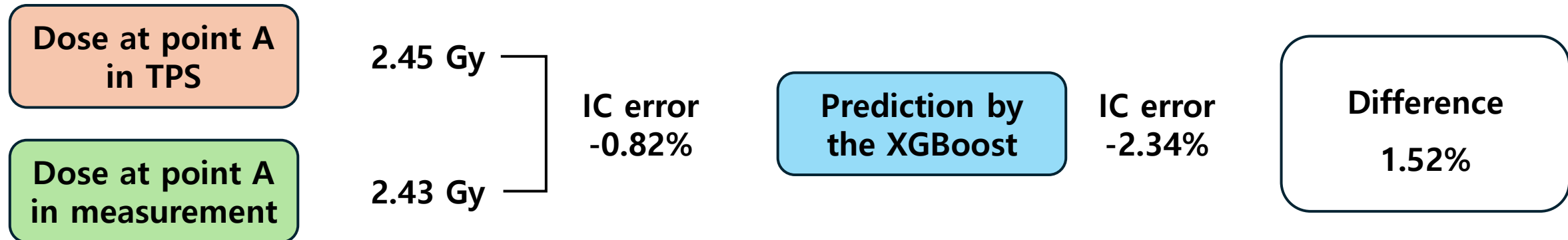
Schematic of a XGboost model

Methods and Materials

- Prediction outcome: Percent differences between the measured ion chamber point dose and the TPS dose



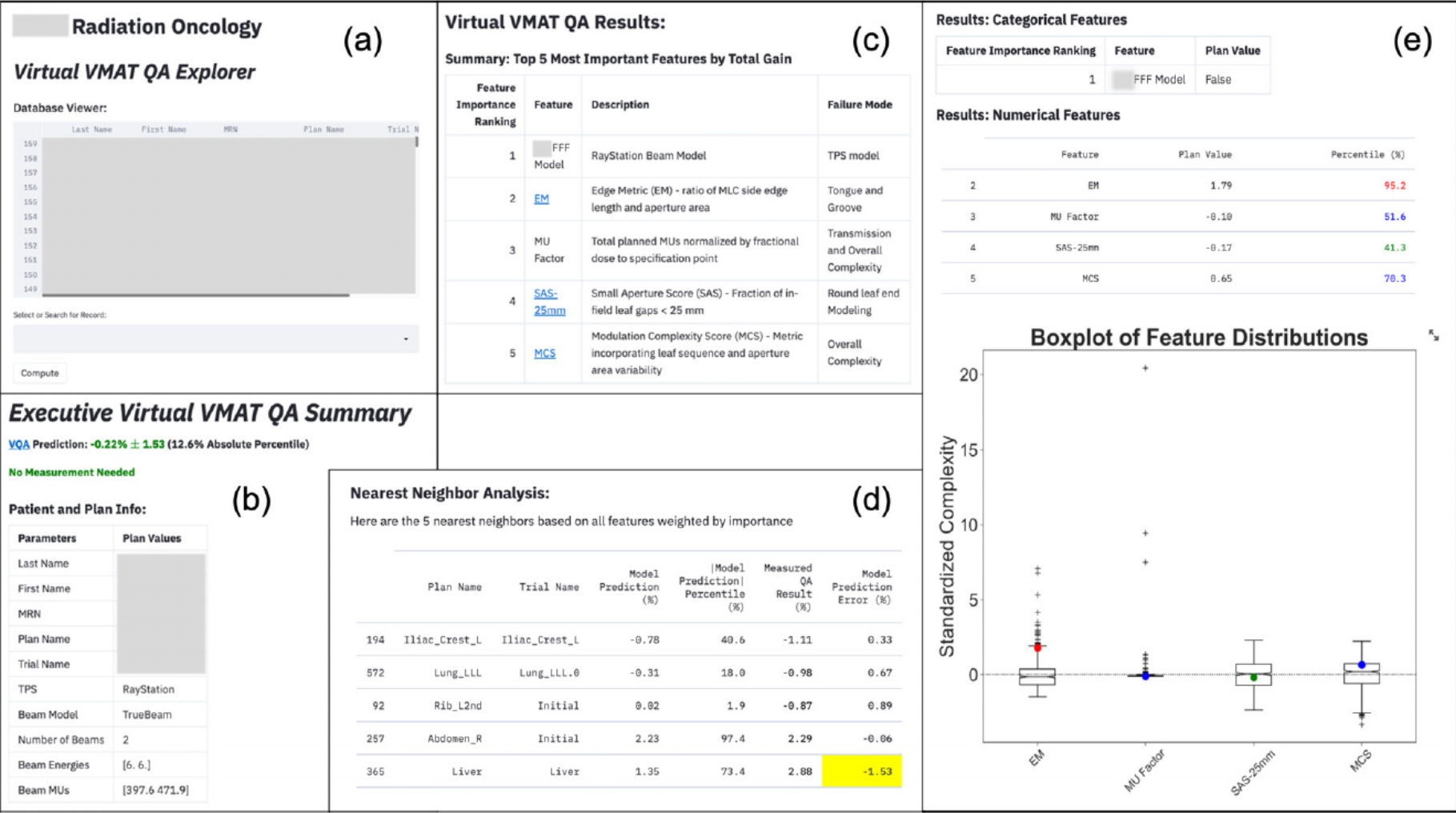
- Virtual QA (VQA) predictions were **prospectively validated over 3 months** at the clinic.





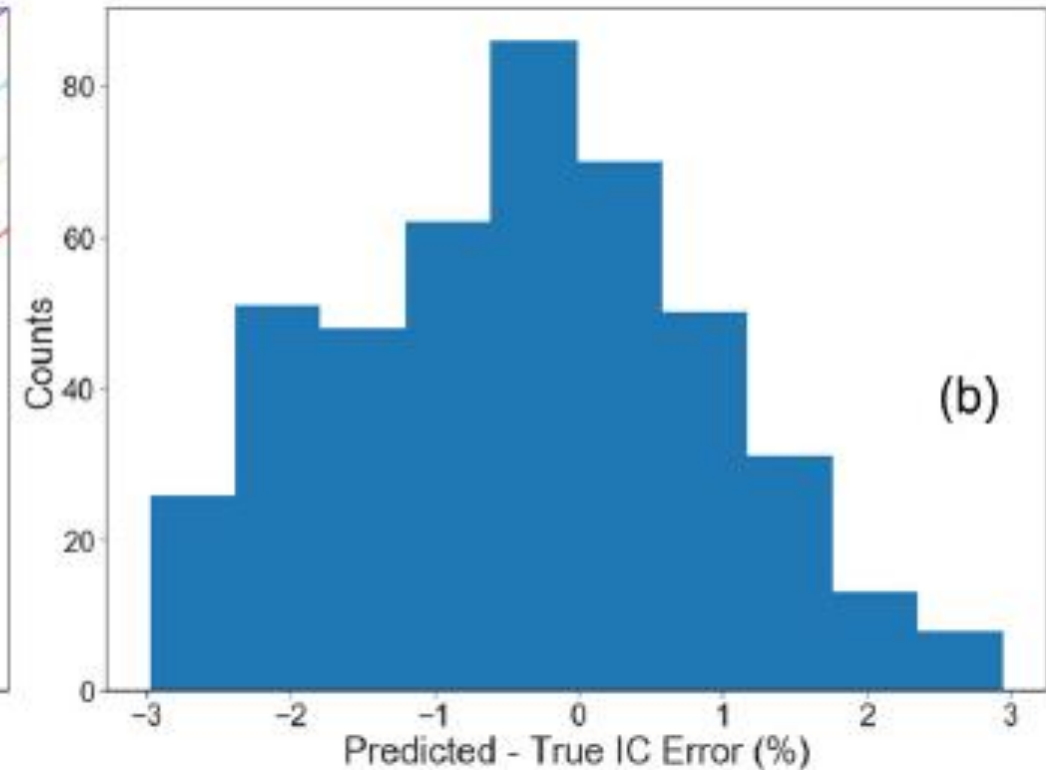
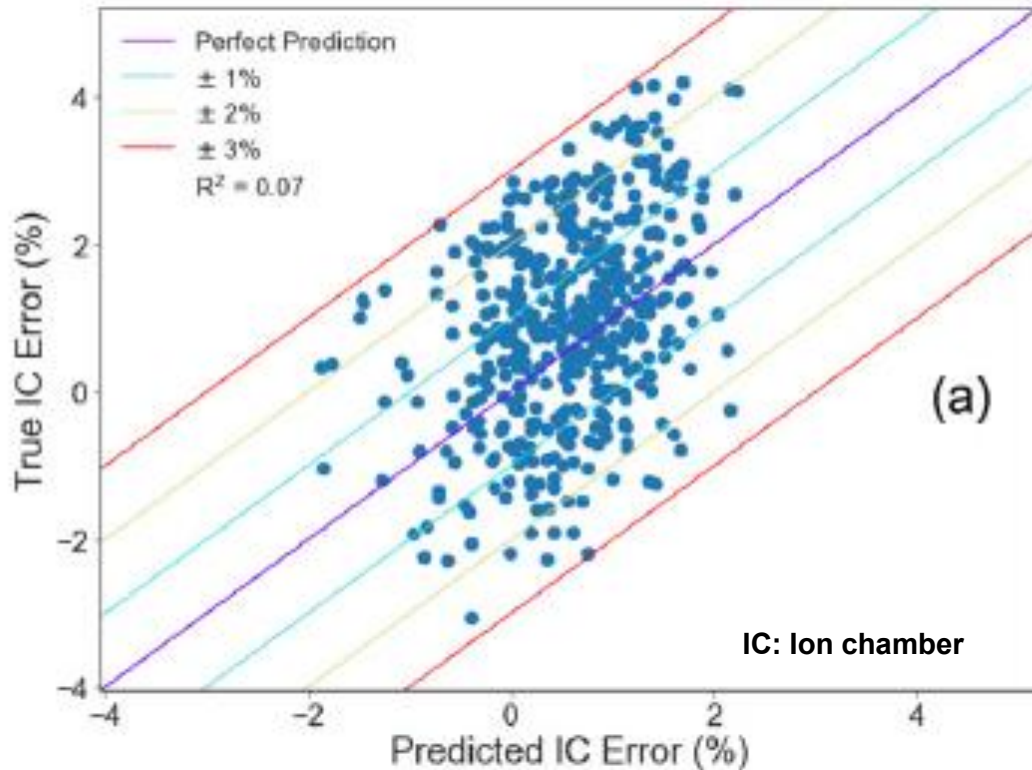
Methods and Materials

- A web application was developed and implemented.



Results

- Over 3 months, 445 VMAT plans were measured, taking 19 minutes per plan.
 - ▶ A mean absolute error of measurement: $1.38 \pm 0.94\%$
- The Virtual QA predictions result
 - ▶ A mean absolute error of **$1.08 \pm 0.77\%$** ($R^2=0.07$)





Results

- A conservative VQA prediction decision threshold was determined to be 1%
 - ▶ plans with absolute predictions < 1% would not require IC measurements

		Sensitivity	Specificity
Clinical pass/fail criterion	3%	81.5%	72.5%
	4%	100%	70%

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

- Approximately 32.5 hours (69.2% of workload) were reduced per month.



Discussion

- A ML models was developed to predict IC disagreements for VMAT PSQA.
- Implemented within an existing clinical workflow for prospective validation with the web-app.
- The XGBoost model was able to predict all 445 prospective clinical PSQA results to within 3% accuracy.
- A conservative 1% VQA threshold would yield a 69.2% reduction in QA workload, saving over 32 hours per month.
- VMAT PSQA rarely results in failing plans given modern TPS modeling and delivery system.
→ potential paradigm-shifting options for implementing VQA safely and effectively.



Thank you for your attention!

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Result

Prediction of EPID image & Gamma analysis

```
test_dataset = CustomDataset(input_data_test, label_data_test, feature_test)
test_loader = DataLoader(test_dataset, batch_size=1, shuffle=False)
#lst_GPR = []
lst_GPR_label = []
lst_GPR_pred = []
lst_GPR_diff = []

dta = 3
abs_dose_diff = 3
thres = 0.1

for test_data in test_loader:
    input_image = test_data['input'] # 입력 이미지
    label_image = test_data['target'] # 입력 이미지
    features = test_data['features']

    input_image, label_image, label_sum, gamma_image_label, GAR_label = gaf.label_gammaAnalysis(generator, input_image,
                                                                                                  label_image, num_features=num_features,
                                                                                                  dta=dta, abs_dose_diff=abs_dose_diff, thres=thres)
    input_image, pred_image, pred_sum, gamma_image_pred, GAR_pred = gaf.pred_gammaAnalysis(generator, input_image,
                                                                                           num_features=features,
                                                                                           dta=dta, abs_dose_diff=abs_dose_diff, thres=thres)
    gaf.plotGAR(input_image.squeeze(), label_sum.squeeze(), gamma_image_label, GAR_label,
                pred_sum.detach().numpy().squeeze(), gamma_image_pred, GAR_pred)

    lst_GPR_label.append(GAR_label[3])
    lst_GPR_pred.append(GAR_pred[3])
    lst_GPR_diff.append(GAR_label[3] - GAR_pred[3])

print(lst_GPR_label)
print(lst_GPR_pred)
print(lst_GPR_diff)
```

Prediction of EPID image & Gamma analysis

